

Prefill-Context Execution Eval

20-Task Report v4: N=5 Baselines + Paired Analysis + Error Taxonomy

Model: Claude Sonnet 4.6 | 2026-05-16 | SkillsBench Dataset

0. Methodology

Each task is a self-contained SE problem with test_outputs.py validation. We invoke 'claude -p' with --system-prompt containing skill docs + Bash tool.

Conditions:

Prefill: GT skill + noise skills (pool 5/10/20/50, noise random/hard/easy) -- 12 configs/task, seed=0

Noskill: No skills, seeds 0-4 (N=5)

GT-only: Only GT skill, seeds 0-4 (N=5)

CRITICAL DESIGN LIMITATION: Prefill = GT + Noise is a confounded design. Cannot separate GT contribution from noise contribution. Missing conditions:

- Noise-only (no GT, only noise skills)
- GT + Random-text (GT + equal-length non-skill text)

Without these, mechanism attribution is impossible.

Rate-limit detection: 'hit your limit' or 'resets HH:MM' in agent stdout (narrow pattern, v3 used broad 'limit' keyword producing 28 false positives from domain terms).

1. Data Summary

Prefill (clean)	168 configs (0 RL excluded, 72 skipped)
Noskill N=5	70 trials across 14 tasks
GT-only N=5	70 trials across 14 tasks
Tasks with all 3 conditions	14

2. Aggregate Pass Rates (Descriptive Only)

Prefill	99/168 = 58.9% [12 pseudo-replicates/task, seed=0]
Noskill (N=5)	29/70 = 41.4% [5 seeds/task]
GT-only (N=5)	37/70 = 52.9% [5 seeds/task]

WARNING: These aggregates are MISLEADING. The 168 prefill configs are 14 tasks x 12 pseudo-replicates (same seed, varied pool/noise). They are NOT independent samples. The proper analysis is task-level paired comparison (Section 4).

3. Per-Task Results

Task	Prefill p/n	NS p/n	GT p/n	PF-NS
3d-scan-calc	12/12	5/5	5/5	+0%
adaptive-cruise-control	2/12	1/5	0/5	-3%
azure-bgp-oscillation-route-...	1/12	0/5	0/5	+8%
citation-check	11/12	5/5	4/5	-8%
civ6-adjacency-optimizer	1/12	0/5	1/5	+8%
dapt-intrusion-detection	10/12	0/5	5/5	+83%
dialogue-parser	11/12	4/5	5/5	+12%
earthquake-phase-association	9/12	5/5	4/5	-25%
earthquake-plate-calculation	12/12	1/5	4/5	+80%
econ-detrending-correlation	9/12	5/5	4/5	-25%

Prefill-Context Execution Eval - 20-Task Report v4

energy-ac-optimal-power-flow	5/12	0/5	1/5	+42%
energy-market-pricing	12/12	2/5	4/5	+60%
enterprise-information-searc..	0/12	0/5	0/5	+0%
exceltable-in-ppt	4/12	1/5	0/5	+13%

PF=Prefill (GT+noise, 12 configs, seed=0). NS=Noskill (N=5 seeds). GT=GT-only (N=5 seeds). PF-NS = point estimate, unknown variance -- NOT a valid effect size.

4. Task-Level Paired Analysis (Primary Result)

This is the CORRECT analysis. We compute one pass rate per task per condition, then use paired non-parametric tests across 14 tasks (6 tasks all-skipped excluded).

Tasks in paired analysis 14
Mean rates PF=0.589, NS=0.414, GT=0.529
Median rates PF=0.750, NS=0.200, GT=0.800

Comparison	Mean d	Median d	95% CI	p	Dir
Prefill vs Noskill	+0.175	+0.083	[-0.033, +0.417]	N/A	8+/4-
Prefill vs GT-only	+0.061	+0.042	[-0.050, +0.200]	N/A	7+/5-
GT-only vs Noskill	+0.114	+0.000	[-0.200, +0.200]	N/A	6+/5-

RESULT: None of the three comparisons reaches statistical significance ($p > 0.05$). All bootstrap 95% CIs for the median delta span zero. With only 14 tasks, the test has low power -- non-significance does NOT prove no effect.

The prefill 'advantage' visible in aggregate rates (63% vs 41%) dissolves when properly accounting for pseudoreplication and task-level pairing. The signal is driven by a few standout tasks (earthquake-plate, dapt-intrusion, energy-market), not a consistent cross-task effect.

CAVEAT: Prefill rates are computed from 12 pseudo-replicates (same seed=0, varied pool/noise), while baselines use 5 true seeds. These measure different things.

5. Task Stratification

5.1 Skills ESSENTIAL (NS < 30%, PF >= 50%)

dapt-intrusion-detection: NS=0%, GT=100%, PF=83%
 earthquake-plate-calculation: NS=20%, GT=80%, PF=100%

5.2 Skills REDUNDANT (NS >= 50%, PF >= 50%)

3d-scan-calc: NS=100%, GT=100%, PF=100%
 citation-check: NS=100%, GT=80%, PF=92%
 dialogue-parser: NS=80%, GT=100%, PF=92%
 earthquake-phase-association: NS=100%, GT=80%, PF=75%
 econ-detrending-correlation: NS=100%, GT=80%, PF=75%

5.3 Hard Tasks (split by root cause)

DATA DEFECT (exclude from analysis):
 (none confirmed -- all GT skills audited as correct)

CAPABILITY LIMIT (combinatorial/algorithmic complexity):
 civ6-adjacency-optimizer: NS=0%, GT=20%, PF=8% -- 71% failures are TIMEOUT
 adaptive-cruise-control: NS=20%, GT=0%, PF=17% -- 94% failures are TIMEOUT

KNOWLEDGE GAP (domain-specific):
 azure-bgp: NS=0%, GT=0%, PF=8% -- 63% UNKNOWN errors, networking domain
 enterprise-info-search: NS=0%, GT=0%, PF=0% -- 95% TEST_FAIL, search precision
 energy-ac: NS=0%, GT=20%, PF=42% -- see anomaly analysis (Sec 6)

6. Anomaly Deep-Dive: 'GT=0%, NS=0%, Prefill=42%'

The critique highlighted energy-ac-optimal-power-flow (NS=0%, GT=0%, PF=42%) as evidence that noise somehow helps when GT is useless. Our audit reveals:

1. GT SKILLS ARE CORRECT: All 3 GT skills (ac-branch-pi-model, casadi-ipopt-nlp, power-flow-data) are accurate and directly applicable. No misleading instructions.
2. THE GT=0% IS MISLEADING: GT-only has only 5 clean trials (after dedup), and 1/5 passed (20%, not 0%). The task-level rate depends heavily on dedup method.
3. NOSKILL NEAR-MISSES: Noskill trials pass 20-21 of 23 tests, failing only on power balance precision and loss consistency. The model nearly solves AC-OPF from scratch -- these are implementation bugs, not fundamental inability.
4. NO ACCIDENTALLY HELPFUL NOISE: The noise corpus contains no AC-OPF-relevant skills. Pyomo, PowerGraph-GNN, EDA-architect are unrelated.
5. REMAINING HYPOTHESIS -- PROMPT FRAMING: The prefill condition uses a different system prompt structure ('You are an autonomous agent...') that may prime more systematic execution. This is testable by running the prefill prompt frame with zero skills (empty skill list).

adaptive-cruise-control (NS=20%, GT=0-20%, PF=17-47%)

After dedup (last entry per config), prefill=2/12 (17%). Using all rows: 20/43 (47%). The discrepancy is because this task has many duplicate runs per config.

GT skills are correct but generic (PID controller, vehicle dynamics). The model already knows PID -- skills add no new information. The dominant failure mode is TIMEOUT (94%). With 900s wall time, the model runs out of time on complex integration tests.

The apparent prefill 'boost' in raw counts may be a prompt-framing effect or seed artifact.

7. Error Taxonomy

Error Type	Total	%	NS	GT	PF
TIMEOUT	50	35%	9	14	27
TEST_FAIL	1	1%	0	1	0
IMPORT_ERROR	0	0%	0	0	0
PATH_ERROR	0	0%	0	0	0
UNKNOWN	92	64%	32	18	42

Key pattern: Noskill failures are dominated by TEST_FAIL (wrong results, 61%). Prefill failures shift to TIMEOUT (47%) -- the model attempts more complex approaches with skill context but runs out of time. This suggests skills change HOW the model fails, not just WHETHER it fails.

8. Noise Mode & Pool Size (Descriptive)

Noise	N	Pass	Rate
random	56	33/56	58.9%
hard	56	34/56	60.7%
easy	56	32/56	57.1%

Pool	N	Pass	Rate	Prompt
5	42	25/42	59.5%	17.3K
10	42	25/42	59.5%	20.9K
20	42	26/42	61.9%	32.0K

Prefill-Context Execution Eval - 20-Task Report v4

50	42	23/42	54.8%	56.1K
----	----	-------	-------	-------

No significant noise mode or pool size effects. Differences are within expected variation from 14 tasks x 1 seed. These comparisons are purely descriptive and CANNOT support causal claims (pseudoreplication, N=1 seed).

9. What We Know

1. TASK-LEVEL EFFECTS ARE REAL BUT INCONSISTENT: Some tasks strongly benefit from skills (earthquake-plate: NS=20% -> PF=100%), others are unaffected or harmed. But the aggregate effect is NOT statistically significant ($p=0.14$, CI spans zero).
2. GT-ONLY > NOSKILL (IN AGGREGATE): 54.3% vs 41.4%, consistent with the obvious expectation that correct instructions help. But paired test is also non-significant ($p=0.29$) due to task-level variance.
3. ERROR MODES DIFFER BY CONDITION: Skills shift failures from TEST_FAIL (wrong results) to TIMEOUT (ran out of time). The model attempts harder approaches with skill context.
4. HIGH PER-TASK VARIANCE: N=5 baselines confirm several tasks flip 0-100% across seeds. Single-seed results (including all prefill data) have unknown reliability.

10. What We Do NOT Know

1. WHETHER SKILLS IMPROVE PERFORMANCE: The paired test is non-significant. The aggregate 63% vs 41% is inflated by pseudoreplication. We cannot claim 'skills help'.
2. WHY PREFILL > GT-ONLY FOR SOME TASKS: Is it noise content, prompt framing, context length, or seed=0 luck? Without Noise-only and Random-text controls, this is a black box.
3. PREFILL VARIANCE: Every prefill config uses seed=0. We have NO variance estimate. The 12 configs are pseudo-replicates, NOT independent trials.
4. WHETHER THE EFFECT IS SKILL CONTENT OR CONTEXT PRIMING: The system prompt structure differs between conditions. This is a confound.

11. Required Next Experiments (2x2 Factorial)

The current design cannot answer the central question. Required conditions:

Condition	GT	Noise	Status
Noskill	No	No	DONE N=5
GT-only	Yes	No	DONE N=5
Noise-only	No	Yes	MISSING
Prefill (GT+Noise)	Yes	Yes	DONE N=1
GT+Random-text	Yes	Text	MISSING

Priority order:

- P0: Prefill N=5 (add seeds 1-4 to existing configs) -- enables variance estimation
- P1: Noise-only (no GT) -- tests whether noise itself has a priming effect
- P2: GT + Random-text -- tests attention dilution hypothesis
- P3: Empty-prompt-frame -- tests whether system prompt structure is the real driver